

AI for Research & Science

Three use-cases



Presentation at the NEO-AMELEAS Workshop

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Outline

I) AI for Skeptical People

- Creating Vector Images with TikZ
- Saving Time: Finding a Counterexample

II) AI for Enthusiasts

- GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)

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AI for Skeptical People



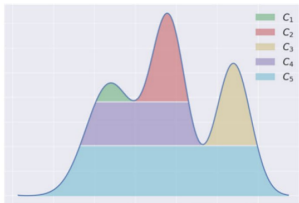
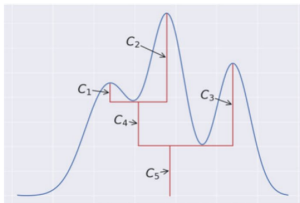


Enhancing Visual Quality: Fast Vector Graphics with AI

- ▶ **Fast & Professional Figures:** AI allows you to generate high-quality illustrations quickly without wasting time.
- ▶ **What is TikZ?:** A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- ▶ **Full Control:** Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.



Example: Target Raster Image

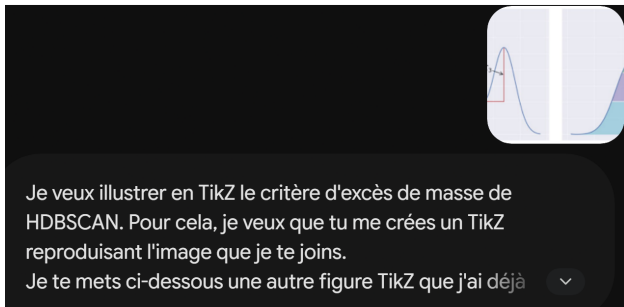


The Challenge:

- ▶ A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- ▶ Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.



Example: Prompting the AI



Human-AI Collaboration:

- ▶ Simply upload the target image and ask the AI to reproduce it in TikZ.
- ▶ Saves hours of calculating coordinates and writing syntax manually.



Example: The Generated TikZ Code

TikZ Code Structure:

```
\begin{tikzpicture}[scale=0.45]
  \tikzset{declare function={f(x) = ...}}

  % Draw density curve
  \draw[thick, blue] plot[domain=0:8] ...;

  % Draw cluster tree bars
  \draw[very thick, red] (1.87,1.70) -- ...;

  % Add labels
  \draw[<-, semithick] (1.87,1.82) -- ...;
\end{tikzpicture}
```

LaTeX Integration:

► Include Package:

- Add `\usepackage{tikz}` in your preamble.

► Direct Placement:

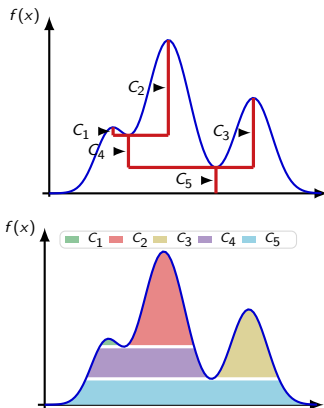
- Paste the generated code directly in the document body.

► No Assets to Manage:

- No need for external image files (PNG/PDF).



Example: The Native TikZ Result



Why it works:

- ▶ Native vector rendering means infinite scaling and crispness.
- ▶ Complete control over font size and styling.

Case Study: A False Mathematical Intuition

The Hypothesis: For the Geometric Stochastic Block Model on a lattice (such as the triangular or square grid), there is no weak recovery for any noise level $p = p_{\text{in}} = 1 - p_{\text{out}} < 1$.

Why the Intuition Felt Correct:

- ▶ **Simulation Artifacts:** Simulations on larger and larger graphs showed the overlap vanishing, which was actually due to the growing mixing time of the MCMC cluster dynamics.
- ▶ **Path Parity Argument:** For any $p < 1$, the number of corrupted edges on a path of length n is $\text{Bin}(n, 1 - p)$. As $n \rightarrow \infty$, the probability of even vs. odd corruptions becomes equal, seemingly washing out all community information.



Case Study: AI Prompt & Response



Réflexion durant 18m 5s >

Il y a un problème assez net avec l'objectif annoncé : sur la grille carrée, l'énoncé "non faiblement recouvrable pour tout $p < 1$ " est faux. Ce n'est pas une petite fissure technique dans le plâtre, c'est plutôt le mur porteur qui vient de déposer sa démission.

The AI Interaction:

- ▶ **The Goal:** Attempting to prove that the recovery threshold $p_c^{(m)} \rightarrow 1$ as the plaquette size $m \rightarrow \infty$ (implying no recovery for any $p < 1$).
- ▶ **The AI's Verdict:** The model spent 18 minutes reasoning and pointed out that the goal itself was false: the MAP estimator achieves weak recovery for p close to 1.

Case Study: Why MAP Works (Peierls Contours)

Proof Sketch by Peierls Argument:

- ▶ **Contours:** Boundaries of incorrect spin decisions form contours γ on the dual grid.
- ▶ **MAP Optimality:** Any contour γ in the MAP estimator must contain at least 50% corrupted edges (otherwise, flipping the region inside would improve the MAP score):

$$B(\gamma) \geq \frac{|\gamma|}{2}$$

- ▶ **Chernoff Bound:** The probability of a fixed contour γ of length ℓ being incorrect is bounded by $(2\sqrt{p(1-p)})^\ell$.
- ▶ **Contour Counting:** The number of contours of length ℓ around any node is at most $\ell 3^\ell$. The probability of an incorrect node is bounded by:

$$\sum_{\ell \geq 4} \ell (6\sqrt{p(1-p)})^\ell \xrightarrow{p \rightarrow 1} 0$$

- ▶ **Conclusion:** Weak recovery is achieved for $p > p_0$, saving months of trying to prove a false statement.

02

AI for Enthusiasts





GitHub, Codespaces, and Antigravity (Gemini-CLI, Google)



Thank you!